

EXECUTIVE PRESENTATION

DIAL-ALERT

A governed path to earlier review of
intradialytic hypotension risk

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Capstone Project, Step 6
September 2026

Decision brief: value, risk, and pilot strategy

Potential value and remaining evidence

Internal test performance supports a governed pilot decision

69.1%

of observed events appear in the
highest-risk 20% of sessions

29.4%

positive predictive value
at 20% review capacity

3.46×

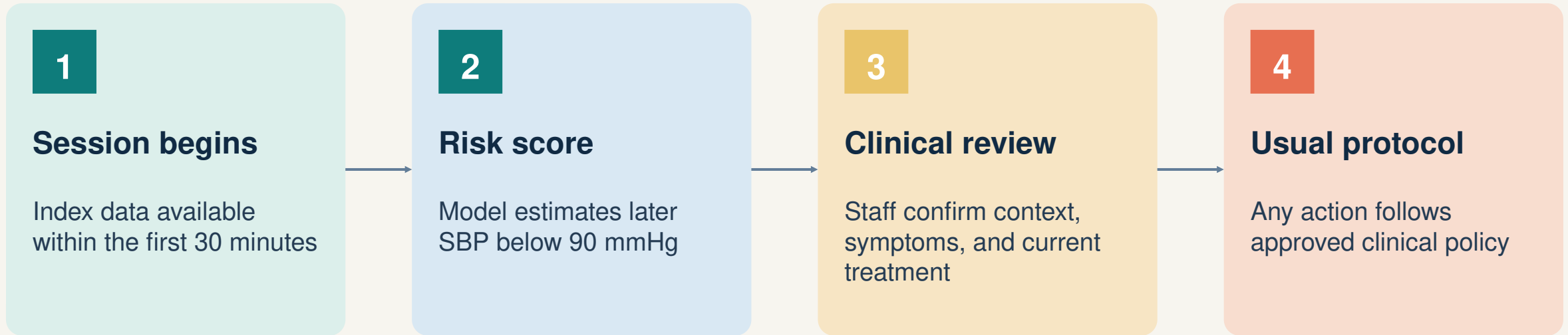
lift over event prevalence

Recommendation

Approve external validation and a prospective silent mode study. Do not use DIAL-ALERT to guide treatment until clinical benefit, workflow safety, and subgroup performance are demonstrated locally.

Proposed workflow for clinician review

The clinician retains responsibility for assessment and action



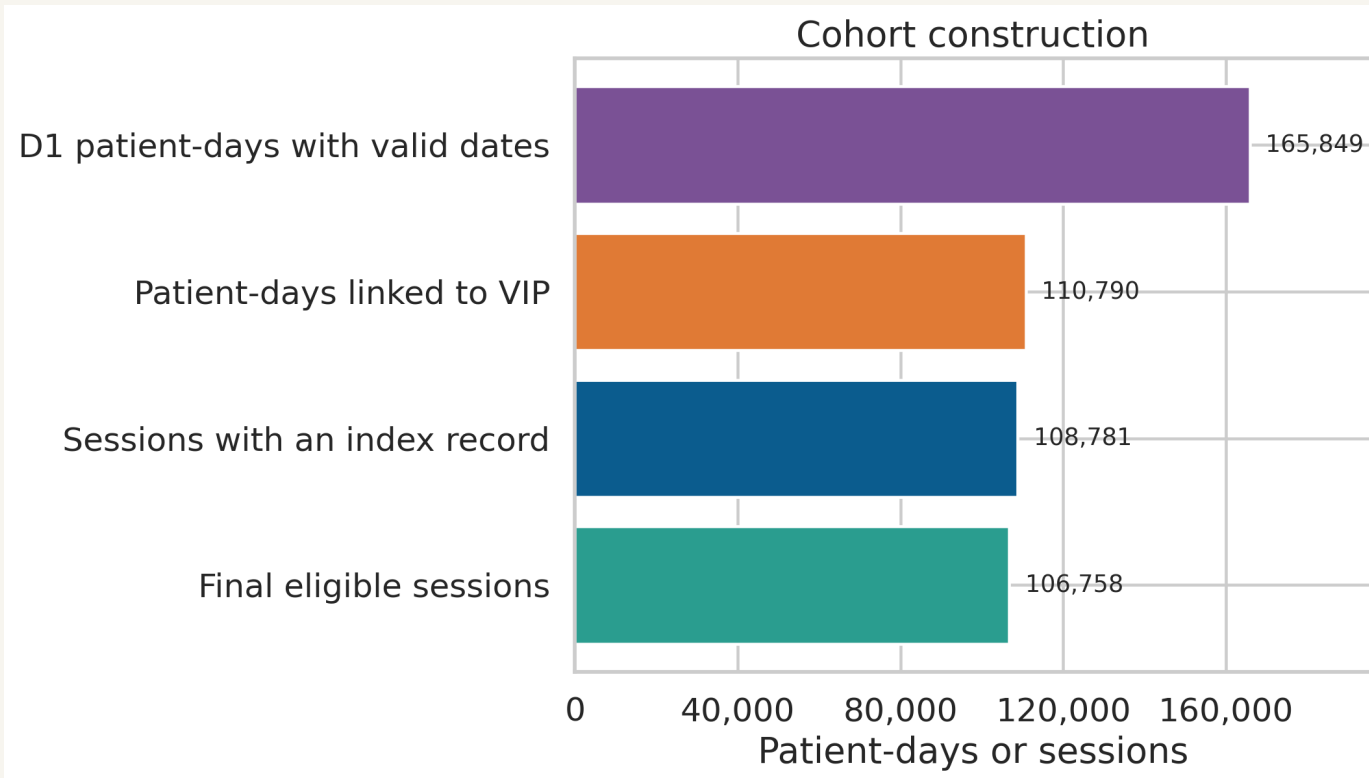
Human controls

Decision support label, clinician override, reason for override, no autonomous prescription, and an audit trail

Proposed workflow for evaluation, not an approved clinical process

Evidence from one retrospective setting

Public HEMOBP data provide strong longitudinal depth from one retrospective source



106,758

eligible sessions

830

unique patients

8.49%

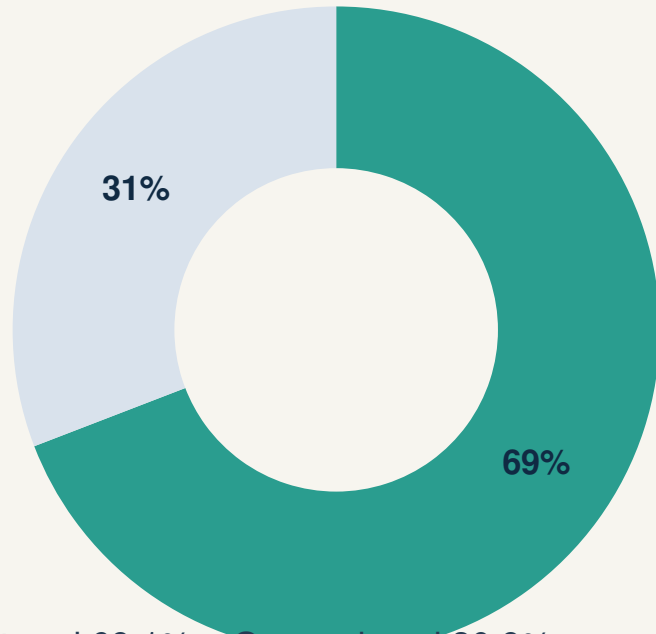
event prevalence

Test set

**21,354 sessions from 170 patients
with zero overlap with development
patients**

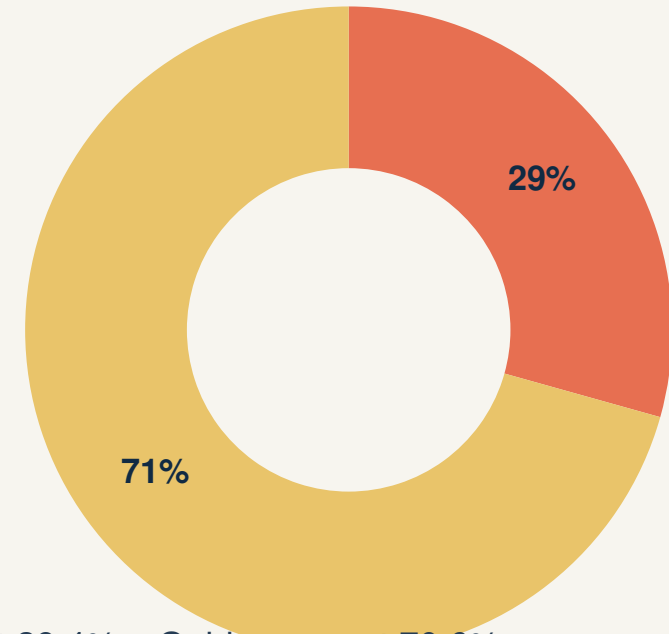
Workload at 20% review capacity

Expected counts per 100 sessions in the internal test population



Teal: captured 69.1% Gray: missed 30.9%

8.5 observed events



Coral: event 29.4% Gold: no event 70.6%

20 alert reviews

These counts describe retrospective concentration of events. They do not estimate events prevented by clinical action.

Calculated from 8.49% prevalence, 69.13% recall, and 20% review capacity

ROI measurement framework

The pilot should estimate benefits and costs before any investment claim

Net value = events avoided × cost of an avoidable event – review labor – intervention cost – integration and monitoring

Benefit evidence

- Change in clinically meaningful IDH events
- Avoided treatment interruption or escalation
- Downstream utilisation linked to the event
- Patient experience and recovery time

Cost evidence

- Minutes of clinical review per alert
- Action rate and unintended intervention burden
- Data integration, training, and support
- Ongoing monitoring, audit, and incident response

Break-even decision: proceed only when measured benefit exceeds the full operating cost at an acceptable safety and equity profile.

Clinical, workload and fairness risks

Fairness varies by metric and several protected attributes are unavailable

Risk register

Missed high-risk session

False negatives remain at every threshold

Alert fatigue

20% capacity produces 14.1 false alerts per 100 sessions

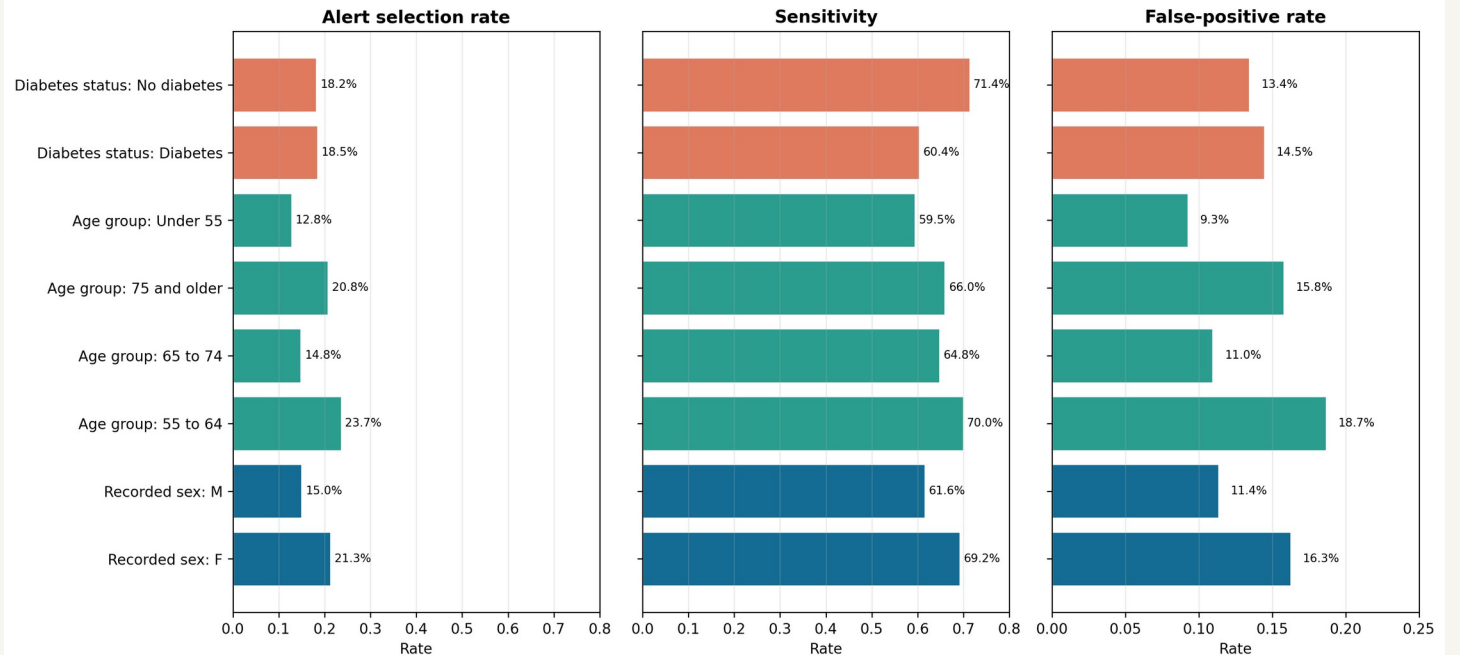
Unequal performance

Recorded sex and age group gaps require review

Automation and privacy

Human override and minimum necessary data are mandatory

Subgroup performance on the patient-disjoint test set



Race, ethnicity, socioeconomic status, and gender identity cannot be audited from the source data.

Governance controls must precede live alerts

Each gate requires documented evidence and an accountable owner

Gate	Evidence required		Accountable owner
1	External validity	Performance and calibration on current local patients	Clinical analytics lead
2	Silent workflow	Alert volume, timing, data quality, and drift without influencing care	Dialysis operations
3	Human factors	Comprehension, overrides, action patterns, and automation bias	Clinical safety lead
4	Equity and privacy	Subgroup errors, broader attributes, access controls, and retention	Governance board
5	Monitoring and rollback	Thresholds for drift, harm, downtime, and model withdrawal	Model owner

Proposed pilot phases and feasibility targets

Progression depends on evidence at the end of each phase

Weeks 0–4

Local validation

Check features, calibration, subgroup errors and workload

Weeks 5–8

Silent evaluation

Score without displaying alerts; time simulated reviews

Weeks 9–12

Conditional supervised study

Proceed only after agreed safety and governance gates

Draft targets for a future fixed-threshold evaluation

Sensitivity $\geq 65\%$, ≤ 20 alerts and ≤ 15 false alerts per 100 sessions

Median active review ≤ 2 min per alert; local approval and uncertainty review required

Decision requested: further validation

The next investment should reduce uncertainty before introducing live alerts

Approve

- A local data and validation partner
- Clinical, informatics, privacy, and equity owners
- A prospective silent mode study
- A predefined go or stop review at 90 days

Stop if

- Performance does not transport locally
- Alert burden exceeds safe staffing capacity
- Subgroup error or calibration gaps are unacceptable
- Clinician behavior creates new safety risk

Until these gates are met, DIAL-ALERT remains an academic decision support prototype and must not determine treatment.